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00001 *****
00002 ** DL Register Area - 120Word
00003 **
00004 ** DW00000 - DW00009 (PLS Use)
00005 ** DW00010 - DW00039 (Bit Use)
00006 ** DW00040 - DW00089 (Word Use)
00007 ** DW00090 - DW00119 (Timer)
00008 *****
00009
00010 "# Program Name : Cutter X, Shift Y Auto Task
00011 "# Function :
00012 "# Call Program :
00013 "# Sub Program :
00014 "# Func Program
00015
00016 VAR;
00017 " Scribe Cutter
00018 00000 LONG IDX_CX = 32;
00019 00001 LONG IDX_CAMX = 33;
00020 00002 LONG AXIS_POS_CAMX = 0;
00021 " Scribe Shift
00022 00003 LONG IDX_CS = 35;
00023 00004 LONG AXIS_POS_CS = 0;
00024 END_VAR;
00025
00026 "Manual/Auto
00027 00005 DB00020 = 0; " Manual Data
00028 00006 IF MB00004 == 1 " AutoRunning
00029 00007 DB00020 = 1; " Recipe Data
00030 00008 ELSE;
00031 00009 IF MB225580 == 1 " [PC Data] Select Data
00032 00010 DB00020 = 1; " Recipe Data
00033 00011 IEND;
00034 00012 IEND;
00035
00036 00013 AXIS_POS_CAMX = IDX_CAMX * 50;
00037 00014 AXIS_POS_CS = IDX_CS * 50;
00038
00039 "Cutter X Command Reset
00040 00015 MW04514 = 0;
00041 00016 MB045150 = 0;
00042 00017 MB045151 = 0;
00043
00044 00018 DW00050 = 0; " Cutter Z Move Command
00045
00046 00019 DB00000 = 0;
00047 00020 WHILE DB00000 == 0 " Cycle Stop Check
00048 "SC Cutter X Unit Command Code Waiting
00049 00021 IOW MW04514 <> 0;
00050 00022 IOW MW04842 == 0;
00051 00023 DW000050 = 0;
00052
00053 "Cutter X Current Position Write
00054 00024 ML16064 = ILA016 " Left Upper Cutter X
00055 00025 ML16066 = ILA096 " Right Upper Cutter
00056
00057 "Cutter Shift Y Current Position Write
00058 00026 ML16070 = ILA196 " Cutter Upper Shift Y
00059
00060 //Cutter Z Waiton Position
00061 00027 IF MW04514 <> 8;
00062 00028 IF MW04514 <> 9;
00063 00029 IF MB045740 | MB045743 == 0;
00064 00030 IOW MW04842 == 0;
00065 00031 MW04842 = 41;
00066 00032 MSEE MPS198;
00067 00033 MW04842 = 0;
00068 00034 IOW MW04842 == 0;
00069 00035 IEND;
00070 00036 IEND;
00071 00037 IEND;
00072
00073 00038 PFORK CX CZ;
00074
00075 CX:
00076
00077 "Scribe Cutter X Motion Type Select
00078 00039 SFORK MW04514 == 1 ? LBL01 " [Cutter X] Waiting Position
00079 MW04514 == 2 ? LBL02 " [Cutter X] Alignment Position
00080 MW04514 == 3 ? LBL03 " [Cutter X] Scribe Start Position
00081 MW04514 == 4 ? LBL04 " [Cutter X] Cutter Z Load Cell Calibration/Measur
ement Position
00082 MW04514 == 5 ? LBL05 " [Cutter X]
00083 MW04514 == 6 ? LBL06 " [Cutter X] Line Camera Check Position
00084 MW04514 == 7 ? LBL07 " [Cutter X] 0 Point Detect Position
00085 MW04514 == 8 ? LBL08 " [Cutter X] Upper Cutter X Wheel Change Position
00086 MW04514 == 9 ? LBL09 " [Cutter X] Not Move(CZ Wait Position)

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00087          MW04514 == 10 ? LBL10          " [Cutter X] Wheel Change Avoid Position
00088          MW04514 == 11 ? LBL11          " [Cutter X] Cutting After Escape Position
00089          MW04514 == 12 ? LBL12          " [Cutter X] Chuck Task End Proce

00090          DEFAULT          ? LBL100;
00091
00092          LBL01:          " [Cutter X] Waiting Position
00093          "[Cutter X] Cutter X Command Execu
00094          IOW MB045150 == 1;
00095
00096          "Cutter X Waiting Pos Write
00097          ML16064 = ML43230          " Left Upper Cutter X
00098          ML16066 = ML43330          " Right Upper Cutter
00099
00100          "Scribe Shift Y Waiting Positi
00101          ML16070 = ML43530          " Scribe Left Upper Shift Y
00102
00103          "Cutter X Pos Moving Permissi
00104          IOW MB45760 & MB4577A == 1;
00105
00106          DW00050 = 41;
00107
00108          MSEE MPS171;          " [Cutter] Scribe Cutter X Move
00109
00110          " Scribe Camera StandBy Position
00111          PLD [LCAMY][LCAMZ][RCAMY][RCAMZ]
00112          MOV [LCAMY]ML43730 [LCAMZ]ML43930 [RCAMY]ML43830 [RCAMZ]ML44030;
00113
00114          "[Cutter X] Auto Complete
00115          MB045151 = 1;          " Scribe Scribe Cutter Command Complete
00116          Bit ON          IOW MB045150 == 0          " Scribe Scribe Cutter Command Execute B
00117          it ON          MB045151 = 0;          " Scribe Scribe Cutter Command Complete
00118          Bit ON
00119          "Cutter X Task Code Reset          " Scribe Scribe Cutter Command Code Sett
00120          ing          MW04514 = 0;
00121
00122          JOINTO LBL0X;
00123
00124          LBL02:          " [Cutter X] Alignment Position
00125          "[Cutter X] Cutter X Command Execu
00126          IOW MB045150 == 1;
00127
00128          MSEE MPS162;          " Alignment Camera Data Calc
00129
00130          ML16070 = ML43554;
00131          IOW MB045761 & MB04577B & MB04577E == 1;
00132
00133          DW00050 = 41;
00134
00135          MSEE MPS171;          " [Cutter] Scribe Cutter X Move
00136
00137          "[Cutter X] Auto Complete
00138          MB045151 = 1;          " Scribe Scribe Cutter Command Complete
00139          Bit ON          IOW MB045150 == 0          " Scribe Scribe Cutter Command Execute B
00140          it ON          MB045151 = 0;          " Scribe Scribe Cutter Command Complete
00141          Bit ON
00142          "Cutter X Task Code Reset          " Scribe Scribe Cutter Command Code Sett
00143          ing          MW04514 = 0;
00144
00145          JOINTO LBL0X;
00146
00147          LBL03:          " [Cutter X] Scribe Start Position
00148          "[Cutter X] Cutter X Command Execu
00149          IOW MB045150 == 1;
00150
00151          IF DB00020 == 1          " Recipe Data          " Scribe
00152          IF ("MB047351 | "MB047350) & !MB15200D ==
00153          Run Check)
00154          DL00040 = ML04914          " Upper Cutter X
00155          DL00042 = ML04916          " Lower Cutter X
00156          ELSE;
00157          DL00040 = ML15202          " Upper Cutter X
00158          DL00042 = ML15204          " Lower Cutter X
00159          IEND;
00160
00161          " Cutter X Scribe Position Setting
00162          IF MB15200D == 1          " JustCutMode 0:
00163          ML16064 = DL00042          " Upper Cutter X
00164          ELSE;
00165          IF ML01142 == 0          " Upper Roller Use(0:Cut
00166          ter 1:Roller)

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00163 00081          ML16064 = DL00040          " Upper Cutter X
00164 00082          ELSE;
00165 00083          IF CW00001 == 1          " Scribe Direction (0: F
orward / 1: Reverse
00166 00084          ML16064 = DL00040 + ML26346          " Upper Cutter :
00167 00085          ELSE;
00168 00086          ML16064 = DL00040 - ML26346          " Upper Cutter X
00169 00087          IEND;
00170 00088          IEND;
00171 00089          IEND;
00172
00173 00090          ML16066 = ML40030[AXIS_POS_CAMX]          " Upper Camera :
00174
00175          " Scribe Shift Y Scribe Start Positi
00176 00091          IF MB15200D == 1          " JustCutMode On
00177 00092          ML16070 = ML40030[AXIS_POS_CS] + ML15220 + GL215(          " Scribe Left Upper Shift
t Y
00178 00093          ELSE;
00179 00094          IF MB047350 == 1          " Scribe Emergency Stop
00180 00095          ML16070 = ML04906          " Scribe Left Upper Shift
t Y
00181 00096          ELSE;
00182 00097          ML16070 = ML40030[AXIS_POS_CS] + ML1521          " Scribe Left Upper Shift
t Y
00183 00098          IEND;
00184 00099          IEND;
00185 00100          ELSE;
00186          " Cutter X Scribe Position Setting
00187 00101          IF ML01142 == 0          " Upper Roller Use(0:Cutter 1:Ro
ller)
00188 00102          ML16064 = ML15202          " Left Upper Cutter X
00189 00103          ELSE;
00190 00104          IF CW00001 == 1          " Scribe Direction (0: Forward /
1: Reverse)
00191 00105          ML16064 = ML15202 + ML26346          " Upper Cutter :
00192 00106          ELSE;
00193 00107          ML16064 = ML15202 - ML26346          " Upper Cutter X
00194 00108          IEND;
00195 00109          IEND;
00196
00197 00110          ML16066 = ML40030[AXIS_POS_CAMX]          " Right Upper Cutter
00198
00199          " Scribe Shift Y Scribe Start Position
00200 00111          ML16070 = ML40030[AXIS_POS_CS] + ML1521          " Scribe Left Upper Shift Y
00201 00112          IEND;
00202
00203 00113          IOW MB045766 & MB04577A & MB04577E == 1;
00204
00205 00114          DW000050 = 43;
00206
00207 00115          MSEE MPS171;          " [Cutter] Scribe Cutter X Move
00208
00209          " [Cutter X] Auto Complete
00210 00116          MB045151 = 1;          " Scribe Scribe Cutter Command C
omplete Bit ON
00211 00117          IOW MB045150 == 0          " Scribe Scribe Cutter Command E
xecute Bit ON
00212 00118          MB045151 = 0;          " Scribe Scribe Cutter Command C
omplete Bit OI
00213          " Cutter X Task Code Reset
00214 00119          MW04514 = 0;          " Scribe Scribe Cutter Command C
ode Setting
00215
00216 00120          JOINTO LBL0X;
00217
00218 00121          LBL04:          " [Cutter X] Cutter Z Load Calibration/Masurement Posit on
00219          " [SC CUTTER] Cutter X Command Execu
00220 00122          IOW MB045150 == 1;
00221
00222          " Cutter X Waiting Pos Write
00223 00123          ML16064 = ML43242          " Left Upper Cutter X
00224 00124          ML16066 = ML43330          " Right Upper Cutter
00225
00226          " Cutter Shift Y Load Calibration/Masurement Write
00227 00125          ML16070 = ML43542;          " Scribe Left Upper Cutter Shift
Y
00228
00229          " CX, CS Moving Permission
00230 00126          IOW MB045763 & MB04577E == 1
00231
00232 00127          DW000050 = 41;
00233
00234 00128          MSEE MPS171;          " [Cutter] Scribe Cutter X Move
00235

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00236          " [Cutter X] Auto Complete
00237 00129      MB045151 = 1;          " Scribe Scribe Cutter Command C
omplete Bit ON
00238 00130      IOW MB045150 == 0      " Scribe Scribe Cutter Command E
xecute Bit ON
00239 00131      MB045151 = 0;          " Scribe Scribe Cutter Command C
omplete Bit ON
00240          " Cutter X Task Code Reset
00241 00132      MW04514 = 0;          " Scribe Scribe Cutter Command C
ode Setting

00242
00243 00133      JOINTO LBL0X;
00244
00245 00134      LBL05:          " [Cutter X] Roller Z Load Calibration/Measurement Posit on
00246          " [Cutter X] Cutter X Command Execu
00247 00135      IOW MB045150 == 1;
00248
00249          " Cutter X Waiting Pos Write
00250 00136      IF CW00001 == 1          " Scribe Direction (0: Forward /
1: Reverse)
00251 00137          ML16064 = ML43242 + ML26346      " Upper Cutter :
00252 00138      ELSE;
00253 00139          ML16064 = ML43242 - ML26346      " Upper Cutter X
00254 00140      IEND;
00255 00141      ML16066 = ML43330          " Right Upper Cutter

00256
00257          " Cutter Shift Y Waiting Pos Write
00258 00142      ML16070 = ML43542;          " Scribe Left Upper Cutter Shift
Y

00259
00260          " LUCX, LLCX, LUCS, LLCS Moving Permission
00261 00143      IOW MB045763 & MB04577E == 1
00262
00263 00144      DW000050 = 41;
00264
00265 00145      MSEE MPS171;          " [Cutter] Scribe Cutter X Move
00266
00267          " [SC CUTTER] Auto Complete
00268 00146      MB045151 = 1;          " Scribe Scribe Cutter Command C
omplete Bit ON
00269 00147      IOW MB045150 == 0      " Scribe Scribe Cutter Command E
xecute Bit ON
00270 00148      MB045151 = 0;          " Scribe Scribe Cutter Command C
omplete Bit ON
00271          " Cutter X Task Code Reset
00272 00149      MW04514 = 0;          " Scribe Scribe Cutter Command C
ode Setting

00273
00274 00150      JOINTO LBL0X;
00275
00276 00151      LBL06:          " [Cutter X] Line Camera Check Positi n
00277          " [SC CUTTER] Cutter X Command Execu
00278 00152      IOW MB045150 == 1;
00279
00280 00153      DW000020 = MW04862;
00281
00282 00154      DB00000010 = 0;
00283 00155      IF GL21508 == 0;
00284 00156          IF MB15301B == 1,          " Just Cut Mode Use
00285 00157              DB00000010 = 1,          "
00286 00158              IEND;
00287 00159      IEND;
00288
00289 00160      SFORK      DW000020 == 13 ? LBL13      " Scribe Start Position
00290          DW000020 == 14 ? LBL14      " Scribe End Positio
00291          DW000020 == 15 ? LBL15      " Scribe Start Positio
00292          DW000020 == 16 ? LBL16      " Scribe End Positio
00293
00294 00161      LBL13:
00295          " Scribe Start Position
00296 00162      IF CW00001 == 1          " Scribe Direction (0: Forward /
1: Reverse)
00297 00163          ML16064 = ML43230          " Upper Cutter X
00298 00164          ML16066 = ML15202 + ML26372      " Upper Camera X
00299 00165          IF DB00000010 == 1;
00300 00166              ML16074 = ML15220 + ML43730 + ML2626      " Left Upper Camera Y
00301 00167          ELSE;
00302 00168              ML16074 = ML15218 + ML43730 + ML2626      " Left Upper Camera Y
00303 00169          IEND;
00304 00170      ELSE;
00305 00171          ML16064 = ML15202 + ML26302 - ML2637      " Upper Cutter X
00306 00172          ML16066 = ML43330          " Upper Camera :
00307 00173          IF DB00000010 == 1;
00308 00174              ML16074 = ML15220 + ML43730 + ML2626      " Left Upper Camera Y
00309 00175          ELSE;
00310 00176              ML16074 = ML15218 + ML43730 + ML2626      " Left Upper Camera Y

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00311      //                      ML16070 = ML43730 + ML26268      " Left Upper Camera Y
00312 00177                      IEND;
00313 00178                      IEND;
00314 00179                      JOINTO LBL1X;
00315
00316 00180                      LBL14:
00317                      " Scribe End Position
00318 00181                      IF CW00001 == 1      " Scribe Direction (0: Forward /

1: Reverse)
00319 00182                      ML16064 = ML43230      " Upper Cutter X
00320 00183                      ML16066 = ML15210 + ML26372      " Upper Camera :
00321 00184                      IF DB0000010 == 1      " Just Cut Mode Use
00322 00185                      ML16074 = ML15224 + ML43730 + ML2627      " Left Upper Camera
00323 00186                      ELSE;
00324 00187                      ML16074 = ML15222 + ML43730 + ML2627      " Left Upper Camera Y
00325 00188                      IEND;
00326 00189                      ELSE;
00327 00190                      ML16064 = ML15210 + ML26302 - ML2637      " Upper Cutter X
00328 00191                      ML16066 = ML43330      " Upper Camera :
00329 00192                      IF DB0000010 == 1      " Just Cut Mode Use
00330 00193                      ML16074 = ML15224 + ML43730 + ML2627      " Left Upper Camera
00331 00194                      ELSE;
00332 00195                      ML16074 = ML15222 + ML43730 + ML2627      " Left Upper Camera Y
00333      //                      ML16070 = ML43730 + ML2627C      " Left Upper Camera

00334 00196                      IEND;
00335 00197                      IEND;
00336 00198                      JOINTO LBL1X;
00337
00338 00199                      LBL15:
00339
00340                      " Scribe Start Positio
00341 00200                      IF CW00001 == 1      " Scribe Direction (0: Forward /

1: Reverse)
00342 00201                      ML16064 = ML43230      " Upper Cutter X
00343 00202                      ML16066 = ML15204 + ML26372      " Upper Camera :
00344
00345 00203                      ML16074 = ML15220 + ML43730 + ML2626      " Left Upper Camera Y
00346 00204                      ELSE;
00347 00205                      ML16064 = ML15204 + ML26302 - ML2637      " Upper Cutter X
00348 00206                      ML16066 = ML43330      " Upper Camera :
00349
00350      //                      ML16070 = ML15220 + ML43730 + ML2626      " Left Upper Camera Y
00351 00207                      ML16074 = ML15220 + ML43730 + ML2626      " Left Upper Camera Y
00352 00208                      IEND;
00353
00354 00209                      JOINTO LBL1X;
00355
00356 00210                      LBL16:
00357                      " Scribe End Position
00358 00211                      IF CW00001 == 1      " Scribe Direction (0: Forward /

1: Reverse)
00359 00212                      ML16064 = ML43230      " Upper Cutter X
00360 00213                      ML16066 = ML43330      " Upper Camera :
00361
00362 00214                      ML16074 = ML43730      " Left Upper Camera

ELSE;
00363 00215                      ML16064 = ML43230      " Upper Cutter X
00364 00216                      ML16066 = ML43330      " Upper Camera :
00365 00217                      ML16074 = ML43730      " Left Upper Camera Y
00366
00367 00218                      IEND;
00368 00219                      JOINTO LBL1X;
00369
00370 00220                      LBL1X:SJOINT;
00371 00221
00372
00373 00222                      DW000050 = 41;
00374
00375 00223                      MSEE MPS171;      " [Cutter] Scribe Cutter X Move
00376
00377 00224                      MW04862 = 0;
00378
00379                      "[Cutter X] Auto Complete
00380 00225                      MB045151 = 1;      " Scribe Scribe Cutter Command C

omplete Bit ON
00381 00226                      IOW MB045150 == 0      " Scribe Scribe Cutter Command E
xecute Bit ON
00382 00227                      MB045151 = 0;      " Scribe Scribe Cutter Command C
omplete Bit OI
00383                      "Cutter X Task Code Reset
00384 00228                      MW04514 = 0;      " Scribe Scribe Cutter Command C
ode Setting
00385
00386 00229                      JOINTO LBL0X;
00387
00388 00230                      LBL07:      " [Cutter X] 0 Point Detect Positi

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00389          "[Cutter X] Cutter X Command Execu'
00390 00231      IOW MB045150 == 1;
00391
00392          "Scribe Chuck Y 0 Level Search Positi n
00393 00232      IF MB154219 == 0                                " UPCX 0 Level Search [0:Cutter
1:Roller]
00394 00233          ML16064 = ML01210[MW04812]                    " Upper Cutter X Position
00395 00234      ELSE;
00396 00235          IF CW00001 == 1                                " Scribe Direction (0: Forward /
1: Reverse)
00397 00236          ML16064 = ML01210[MW04812] + ML2634          " Upper Cutter X Positio
00398 00237      ELSE;
00399 00238          ML16064 = ML01210[MW04812] - ML2634          " Upper Cutter X Positio
00400 00239      IEND;
00401 00240      IEND;
00402
00403          "Camera X Escape Position
00404 00241      ML16066 = ML43330                                " Upper Camera X Position
00405
00406          " Shift Y 0 Detect Position
00407 00242      ML16070 = ML40030[AXIS_POS_CS]                    " Scribe Upper Shift Y Position
00408
00409 00243      DW00050 = 41;
00410 00244      IF !MB000004 == 1;
00411 00245          MW04842 = 41;
00412 00246          MSEE MPS198;
00413 00247          MW04842 = 0;
00414 00248      ELSE;
00415 00249          DW00050 = 41;
00416 00250      IEND;
00417
00418          "LUCX, LLCX, LUCS, LLCS 0 Detect Position Moving Permission
00419 00251      IOW MB045765 & MB04577E"MB04576D" == 1;
00420
00421 00252      MSEE MPS171;                                " [Cutter] Scribe Cutter X Move
00422
00423          "[Cutter X] Auto Complete
00424 00253      MB045151 = 1;                                " Scribe Scribe Cutter Command C
omplete Bit ON
00425 00254      IOW MB045150 == 0                                " Scribe Scribe Cutter Command E
xecute Bit ON
00426 00255      MB045151 = 0;                                " Scribe Scribe Cutter Command C
omplete Bit OI
00427          "Cutter X Task Code Reset
00428 00256      MW04514 = 0;                                " Scribe Scribe Cutter Command C
ode Setting
00429
00430 00257      JOINTO LBL0X;
00431
00432 00258      LBL08:      " [Cutter X] Upper Cutter X Wheel Change Position
00433          "[Cutter X] Cutter X Command Execu'
00434 00259      IOW MB045150 == 1;
00435
00436          "Left Upper Cutter X Wheel Change Position
00437 00260      ML16064 = ML43258;
00438          "Right Upper Cutter X Wheel Escape Positi
00439 00261      ML16066 = ML43330
00440
00441          "Cutter Left Upper Shift Wheel Change Position
00442 00262      ML16070 = ML43562;
00443
00444          "Upper Cutter X Wheel Change Position Moving Permiss:
00445      " IOW MB45767 & MB4576B & MB04577A == 1;
00446 00263      IOW MB04577A == 1;
00447
00448 00264      MSEE MPS171;                                " [Cutter] Scribe Cutter X Move
00449
00450          "Upper Cutter X Wheel Change Position Che
00451 00265      IOW MB46008 == 1;
00452
00453          "[Cutter X] Auto Complete
00454 00266      MB045151 = 1;                                " Scribe Scribe Cutter Command C
omplete Bit ON
00455 00267      IOW MB045150 == 0                                " Scribe Scribe Cutter Command E
xecute Bit ON
00456 00268      MB045151 = 0;                                " Scribe Scribe Cutter Command C
omplete Bit OI
00457          "Cutter X Task Code Reset
00458 00269      MW04514 = 0;                                " Scribe Scribe Cutter Command C
ode Setting
00459
00460 00270      JOINTO LBL0X;
00461
00462 00271      LBL09:
00463 00272      IF CW00098 == 1;
00464          "[Cutter X] Cutter X Command Execu'
00465 00273      IOW MB045150 == 1;

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00466
00467 00274          IF ML05310 | ML05810 == 1;
00468 00275              DW00050 = 46;
00469 00276          ELSE;
00470 00277              DW00050 = 41;
00471 00278          IEND;
00472
00473          "[Cutter X] Auto Complete
00474 00279          MB045151 = 1;          " Scribe Scribe Cutter Comma
nd Complete Bit ON
00475 00280          IOW MB045150 == 0          " Scribe Scribe Cutter Comma
nd Execute Bit ON
00476 00281          MB045151 = 0;          " Scribe Scribe Cutter Comma
nd Complete Bit O
00477          "Cutter X Task Code Reset
00478 00282          MW04514 = 0;
00479 00283          IEND;
00480
00481 00284          JOINTO LBL0X;
00482
00483 00285          LBL10:          "[Cutter X] Wheel Change Avoid Position
00484          "[Cutter X] Cutter X Command Execu'
00485 00286          IOW MB045150 == 1;
00486
00487          "Cutter X Waiting Pos Write
00488 00287          ML16064 = ML43262          " Left Upper Cutter X
00489 00288          ML16066 = ML43330          " Right Upper Cutter

00490
00491          "Scribe Shift Y Waiting Positio
00492 00289          ML16070 = ML43530          " Scribe Left Upper Shift Y
00493
00494 00290          DW00050 = 41;
00495
00496 00291          MSEE MPS171;          " [Cutter] Scribe Cutter X Move
00497
00498          "[Cutter X] Auto Complete
00499 00292          MB045151 = 1;          " Scribe Scribe Cutter Command C
omplete Bit ON
00500 00293          IOW MB045150 == 0          " Scribe Scribe Cutter Command E
xecute Bit ON
00501 00294          MB045151 = 0;          " Scribe Scribe Cutter Command C
omplete Bit OI
00502          "Cutter X Task Code Reset
00503 00295          MW04514 = 0;          " Scribe Scribe Cutter Command C
ode Setting
00504
00505 00296          JOINTO LBL0X;
00506
00507 00297          LBL11:          "[Cutter X] Cutting After Escape Position
00508          "[Cutter X] Cutter X Command Execu'
00509 00298          IOW MB045150 == 1;
00510
00511          "Cutter X Waiting Pos Write
00512 00299          ML16064 = ML43270          " Left Upper Cutter X
00513 00300          ML16066 = ML43330          " Right Upper Cutter X
00514
00515          "Scribe Shift Y Waiting Positio
00516 00301          ML16070 = ML43530          " Scribe Left Upper Shift Y
00517
00518          "Cutter X Pos Moving Permissio
00519 00302          IOW MB45760 & MB4577A == 1;
00520
00521 00303          DW00050 = 41;
00522
00523 00304          MSEE MPS171;          " [Cutter] Scribe Cutter X Move
00524
00525          "[Cutter X] Auto Complete
00526 00305          MB045151 = 1;          " Scribe Scribe Cutter Command C
omplete Bit ON
00527 00306          IOW MB045150 == 0          " Scribe Scribe Cutter Command E
xecute Bit ON
00528 00307          MB045151 = 0;          " Scribe Scribe Cutter Command C
omplete Bit OI
00529          "Cutter X Task Code Reset
00530 00308          MW04514 = 0;          " Scribe Scribe Cutter Command C
ode Setting
00531
00532 00309          JOINTO LBL0X;
00533
00534 00310          LBL12:
00535          "[Cutter X] Auto Complete Reset
00536 00311          MB045150 = 0;          " Scribe Scribe Cutter Command C
omplete Bit ON
00537 00312          MB045151 = 0;          " Scribe Scribe Cutter Command C
omplete Bit OI
00538          "Cutter X Task Code Reset

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00539 00313          MW04514 = 0;          " Scribe Scribe Cutter Command C
ode Setting
00540
00541 00314          JOINTO LBL0X;
00542
00543 00315          LBL100:
00544 00316          JOINTO LBL0X;
00545 00317          LBL0X: SJOINT;
00546
00547          "Scribe Cutter X Motion Type Select
00548 00318          SFORK  MW04514 == 17 ? LBL17          " [Cutter X]
00549          MW04514 == 18 ? LBL18          " [Cutter X]
00550          MW04514 == 19 ? LBL19          " [Cutter X]
00551          MW04514 == 20 ? LBL20          " [Cutter X]
00552          MW04514 == 21 ? LBL21          " [Cutter X] DummyCut Alignment Position
00553          MW04514 == 22 ? LBL22          " [Cutter X] Cutter Shift Cutting After Escape Pos
ition (Chuck)
00554          MW04514 == 23 ? LBL23          " [Cutter X] Cutter Shift Cutting After Escape Pos
ition (Table)
00555          DEFAULT ? LBL102;
00556
00557
00558 00319          LBL17:          " [Cutter X] Upper Cutter X Wheel Change Align Start Position (Unuse)
00559          "[Cutter X] Cutter X Command Execu'
00560 00320          IOW MB045150 == 1;
00561
00562          "Cutter X Waiting Pos Write
00563 00321          ML16064 = ML43270          " Left Upper Cutter X
00564 00322          ML16066 = ML43330          " Right Upper Cutter X
00565
00566          "Scribe Shift Y Waiting Positi
00567 00323          ML16070 = ML43558          " Scribe Left Upper Shift Y
00568
00569          "Cutter X Pos Moving Permissi
00570 00324          IOW MB45760 & MB45768 & MB4577A == 1;
00571
00572 00325          DW00050 = 41;
00573
00574 00326          MSEE MPS171;          " [Cutter] Scribe Cutter X Move
00575
00576          "[Cutter X] Auto Complete
00577 00327          MB045151 = 1;          " Scribe Scribe Cutter Command Complete
Bit ON
00578 00328          IOW MB045150 == 0          " Scribe Scribe Cutter Command Execute B
it ON
00579 00329          MB045151 = 0;          " Scribe Scribe Cutter Command Complete
Bit ON
00580          "Cutter X Task Code Reset
00581 00330          MW04514 = 0;          " Scribe Scribe Cutter Command Code Sett
ing
00582
00583 00331          JOINTO LBL2X;
00584
00585 00332          LBL18:          " [Cutter X] Upper Cutter X Wheel Change Align End Position (Unuse)
00586          "[Cutter X] Cutter X Command Execu'
00587 00333          IOW MB045150 == 1;
00588
00589          "Cutter X Waiting Pos Write
00590 00334          ML16064 = ML43274          " Left Upper Cutter X
00591 00335          ML16066 = ML43330          " Right Upper Cutter X
00592
00593          "Scribe Shift Y Waiting Positi
00594 00336          ML16070 = ML43562          " Scribe Left Upper Shift Y
00595
00596          "Cutter X Pos Moving Permissi
00597 00337          IOW MB45760 & MB45768 & MB4577A == 1;
00598
00599 00338          DW00050 = 41;
00600
00601 00339          MSEE MPS171;          " [Cutter] Scribe Cutter X Move
00602
00603          "[Cutter X] Auto Complete
00604 00340          MB045151 = 1;          " Scribe Scribe Cutter Command Complete
Bit ON
00605 00341          IOW MB045150 == 0          " Scribe Scribe Cutter Command Execute B
it ON
00606 00342          MB045151 = 0;          " Scribe Scribe Cutter Command Complete
Bit ON
00607          "Cutter X Task Code Reset
00608 00343          MW04514 = 0;          " Scribe Scribe Cutter Command Code Sett
ing
00609
00610 00344          JOINTO LBL2X;
00611
00612 00345          LBL19:          " [Cutter X] Upper Cutter X Wheel Change Align Start Position (Unuse)
00613          "[Cutter X] Cutter X Command Execu'
00614 00346          IOW MB045150 == 1;

```

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00615
00616 "Cutter X Waiting Pos Write
00617 00347 ML16064 = ML43262 " Left Upper Cutter X
00618 00348 ML16066 = ML43330 " Right Upper Cutter X
00619
00620 "Scribe Shift Y Waiting Positio
00621 00349 ML16070 = ML43530 " Scribe Left Upper Shift Y
00622
00623 "Cutter X Pos Moving Permissio
00624 00350 IOW MB45760 & MB45768 & MB4577A == 1;
00625
00626 00351 DW000050 = 41;
00627
00628 00352 MSEE MPS171; " [Cutter] Scribe Cutter X Move
00629
00630 "[Cutter X] Auto Complete
00631 00353 MB045151 = 1; " Scribe Scribe Cutter Command Complete
00632 00354 Bit ON IOW MB045150 == 0 " Scribe Scribe Cutter Command Execute B
00633 00355 it ON MB045151 = 0; " Scribe Scribe Cutter Command Complete
00634 Bit ON
00635 00356 "Cutter X Task Code Reset " Scribe Scribe Cutter Command Code Sett
ing MW04514 = 0;
00636
00637 00357 JOINTO LBL2X;
00638
00639 00358 LBL20: " [Cutter X] Lower Cutter X Wheel Change Align End Posit (Unuse)
00640 "[Cutter X] Cutter X Command Execu
00641 00359 IOW MB045150 == 1;
00642
00643 "Cutter X Waiting Pos Write
00644 00360 ML16064 = ML43262 " Left Upper Cutter X
00645 00361 ML16066 = ML43530 " Right Upper Cutter X
00646
00647 "Scribe Shift Y Waiting Positio
00648 00362 ML16070 = ML43530 " Scribe Left Upper Shift Y
00649
00650 "Cutter X Pos Moving Permissio
00651 00363 IOW MB45760 & MB45768 & MB4577A == 1;
00652
00653 00364 DW000050 = 41;
00654
00655 00365 MSEE MPS171; " [Cutter] Scribe Cutter X Move
00656
00657 "[Cutter X] Auto Complete
00658 00366 MB045151 = 1; " Scribe Scribe Cutter Command Complete
00659 00367 Bit ON IOW MB045150 == 0 " Scribe Scribe Cutter Command Execute B
00660 00368 it ON MB045151 = 0; " Scribe Scribe Cutter Command Complete
00661 Bit ON
00662 00369 "Cutter X Task Code Reset " Scribe Scribe Cutter Command Code Sett
ing MW04514 = 0;
00663
00664 00370 JOINTO LBL2X;
00665
00666 00371 LBL21: " [Cutter X] DummyCut Alignment Position
00667 "[Cutter X] Cutter X Command Execu
00668 00372 IOW MB045150 == 1;
00669
00670 00373 IF CW00001 == 1 " Scribe Direction (0: Forward / 1: Reve
rse)
00671 00374 ML16066 = ML01190 " Left Upper Camera Position Setting
00672 00375 ML16064 = ML01192 " Right Upper Camera Position Sett
00673 00376 ELSE;
00674 00377 ML16064 = ML01190 " Left Upper Camera Position Setting
00675 00378 ML16066 = ML01192 " Right Upper Camera Position Setting
00676 00379 IEND;
00677
00678 00380 ML16070 = ML40030[AXIS_POS_CS];
00679
00680 00381 IOW MB045761 & MB04577B & MB04577E == 1;
00681
00682 00382 DW000050 = 41;
00683
00684 00383 MSEE MPS171; " [Cutter] Scribe Cutter X Move
00685
00686 "[Cutter X] Auto Complete
00687 00384 MB045151 = 1; " Scribe Scribe Cutter Command Complete
00688 00385 Bit ON IOW MB045150 == 0 " Scribe Scribe Cutter Command Execute B
00689 00386 it ON MB045151 = 0; " Scribe Scribe Cutter Command Complete

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```

Bit ON
00690          "Cutter X Task Code Reset
00691 00387      MW04514 = 0;          " Scribe Scribe Cutter Command Code Sett
ing
00692
00693 00388      JOINTO LBL2X;
00694
00695 00389      LBL22:          " [Cutter X] Cutter Shift Cutting After Escape Position (Chuck)
00696          " [SC← CUTTER] Cutter X Command Execute
00697 00390          IOW MB045150 == 1;
00698
00699 00391          IF ML05308 == 1;
00700 00392              IF MB045745 == 1          " CX Escape Position Move
00701                  " Cutter X Waiting Pos Write
00702 00393                  ML16064 = ML43270      " Left Upper Cutter X
00703 00394                  ML16066 = ML43330      " Right Upper Cutter

00704 00395          ELSE;
00705 00396              ML16064 = ML43230          " Left Upper Cutter X
00706 00397              ML16066 = ML43330      " Right Upper Cutter

00707 00398          IEND;
00708 00399      ELSE;
00709          " Cutter X Feedback Pos Write
00710 00400          ML16064 = ILA016          " Left Upper Cutter X
00711 00401          ML16066 = ILA096      " Right Upper Cutter
00712 00402      IEND;
00713
00714          "→ Scribe Shift Y Escape Position
00715 00403      ML16070 = ML43534          " Scribe Left Upper Shift Y
00716
00717          "→ Cutter X Pos Moving Permission
00718 00404      IOW MB45760 & MB45768 & MB4577A == 1;
00719
00720 00405      DW00050 = 41;
00721
00722 00406      MB04515B = 1;          " Cutter Shift Escape Pos Move(C
huck)
00723 00407      MSEE MPS171;          " [Cutter] Scribe Cutter X Move
00724 00408      MB04515B = 0;          " Cutter Shift Escape Pos Move(C
huck)

00725
00726          "→ [SC← CUTTER] Auto Complete
00727 00409      MB045151 = 1;          " Scribe ← Scribe Cutter Comman
d Complete Bit ON
00728 00410          IOW MB045150 == 0          " Scribe → Scribe Cutter Comman
d Execute Bit ON
00729 00411      MB045151 = 0;          " Scribe ← Scribe Cutter Comman
d Complete Bit O
00730          "→ Cutter X Task Code Reset
00731 00412      MW04514 = 0;          " Scribe → Scribe Cutter Comman
d Code Setting

00732
00733 00413      JOINTO LBL2X;
00734
00735 00414      LBL23:          " [Cutter X] Cutter Shift Cutting After Escape Position (Table)
00736          "→ [SC← CUTTER] Cutter X Command Execute
00737 00415          IOW MB045150 == 1;
00738
00739 00416          IF CW00001 == 1          " Scribe Direction (0: Forward / 1: Reve
rse)
00740              "→ Cutter X Feedback Pos Write
00741 00417              ML16064 = ILA016 - 10000      " Left Upper Cutter X
00742 00418              ML16066 = ILA096          " Right Upper Cutter

00743 00419          ELSE;
00744              "→ Cutter X Feedback Pos Write
00745 00420              ML16064 = ILA016 + 10000      " Left Upper Cutter X
00746 00421              ML16066 = ILA096          " Right Upper Cutter
00747 00422          IEND;
00748
00749          "→ Scribe Shift Y Escape Position
00750 00423      ML16070 = ML43534          " Scribe Left Upper Shift Y
00751
00752          "→ Cutter X Pos Moving Permission
00753 00424      IOW MB45760 & MB45768 & MB4577A == 1;
00754
00755 00425      DW00050 = 41;
00756
00757 00426      MB04515C = 1;          " Cutter Shift Escape Pos Move(Table)
00758 00427      MSEE MPS171;          " [Cutter] Scribe Cutter X Move
00759 00428      MB04515C = 0;          " Cutter Shift Escape Pos Move(Table)
00760
00761          "→ [SC← CUTTER] Auto Complete
00762 00429      MB045151 = 1;          " Scribe ← Scribe Cutter Command Comple
te Bit ON
00763 00430          IOW MB045150 == 0          " Scribe → Scribe Cutter Command Execut

```

```
00764 00431 e Bit ON MB045151 = 0; " Scribe ← Scribe Cutter Command Comple
00765 te Bit ON "→ Cutter X Task Code Reset
00766 00432 MW04514 = 0; " Scribe → Scribe Cutter Command Code S
00767 etting
00768 00433 JOINTO LBL2X;
00769
00770 00434 LBL102:
00771 00435 JOINTO LBL2X;
00772 00436 LBL2X: SJOINT;
00773
00774 00437 DW00050 = 99;
00775 00438 JOINTO LabelX;
00776
00777 CZ:
00778 00439 IF MB000004 == 1 " [Move] Scribe Cutter X - Shift
Y
00779 00440 IF !MB00800 & MB00802 & !DB000021 == " Idling & don't Cutter
00780 00441 IOW DW00050 <> 0;
00781 00442 IF DW00050 <> 99;
00782 00443 IOW MW04842 == 0;
00783 00444 MW04842 = DW00050 " Cutter Z Code Write
00784 00445 MSEE MPS198; " Cutter Z Motion
00785 00446 IOW MW04842 == 0;
00786 00447 IEND;
00787 00448 IEND;
00788 00449 IEND;
00789
00790 00450 JOINTO LabelX,
00791 00451 LabelX: PJOINT;
00792
00793 " Scribe Stop Request & [ SC BCHK, TBL] Break Stop Request
00794 00452 IF MB045013 & MB045709 == 1;
00795 00453 DB00000 = 1;
00796 00454 IEND;
00797 00455 EOX;
00798 00456 WEND;
00799
00800 00457 END;
```